

EECS 498-010 Principles of Machine Learning

Linear Algebra Review¹

University of Michigan

Yixuan Jia (Isaac)

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¹Reference: CS229 Linear Algebra Review and Reference

Linear Algebra Review

- Basic Concepts and Notation
- Matrix Multiplication
- Operations and Properties
- Matrix Calculus
- Open questions if time left

Linear Algebra Tricks

- Clear about the notations
- Always check the shapes
- Use toy examples for a trail

$$A, \times \underline{a, b}$$

$$\begin{bmatrix} 1 \end{bmatrix} \in$$

A. shape

$$m \times n \quad n \times p$$

$$C = AB$$

Basic Concepts and Notation

Consider the following system of equations:

$$\begin{aligned}4x_1 - 5x_2 &= -13 \\ -2x_1 + 3x_2 &= 9.\end{aligned}$$

You can find a unique solution for x_1 and x_2 (unless the equations are somehow degenerate). In matrix notation, we can write the system more compactly as

$$Ax = b$$

with

$$A = \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} -13 \\ 9 \end{bmatrix}$$

Basic Concepts and Notation

- By $A \in \mathbb{R}^{m \times n}$ we denote a matrix with m rows and n columns, where the entries of A are real numbers.
- By $x \in \mathbb{R}^n$, we denote a vector with n entries.
 - column vector
 - row vector
- The i th element of a vector x is denoted x_i :

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

3 × 4

A. shape

(3, 4)

x. shape → $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

(3,)

Basic Concepts and Notation

- We use the notation a_{ij} (or A_{ij} , $A_{i,j}$, etc) to denote the entry of A in the i th row and j th column:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Basic Concepts and Notation

- We denote the j th column of A by $\mathbf{a}^j$ or $A_{:,j}$:

$$A = \left[\begin{array}{c|c|c|c} & & & \\ \mathbf{a}^1 & \mathbf{a}^2 & \cdots & \mathbf{a}^n \\ & & & \end{array} \right]$$

- We denote the i th row of A by $\mathbf{a}^T$ or $A_{i,:}$:

$$A = \left[\begin{array}{c} - & \mathbf{a}_1^T & - \\ - & \mathbf{a}_2^T & - \\ & \vdots & \\ - & \mathbf{a}_m^T & - \end{array} \right]$$

Matrix Multiplication

The product of two matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ is the matrix

$$C = AB \in \mathbb{R}^{m \times p},$$

where

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}.$$

Note that in order for the matrix product to exist, the number of columns in A must equal the number of rows in B .

$$\begin{matrix} 2 \times 2 & 2 \times 2 \\ \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} & \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \end{matrix} = \begin{matrix} 2 \\ 2 \end{matrix} \begin{matrix} 3 & 3 \\ 6 & 6 \end{matrix}$$

Matrix Multiplication: Vector-Vector Products

Given two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the quantity $\mathbf{x}^T \mathbf{y}$, sometimes called the inner product or dot product of the vectors, is a real number given by

$$\mathbf{x}^T \mathbf{y} \in \mathbb{R} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i.$$

Observe that inner products are really just special case of matrix multiplication. Note that it is always the case that $\mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x}$.

$$\mathbf{y}^T \mathbf{x} = \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Matrix Multiplication: Vector-Vector Products

Given vectors $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^n$ (not necessarily of the same size), $\mathbf{xy}^T \in \mathbb{R}^{m \times n}$ is called the outer product of the vectors. It is a matrix whose entries are given by $(\mathbf{xy}^T)_{ij} = x_i y_j$, i.e.,

$$\mathbf{xy}^T \in \mathbb{R}^{m \times n} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \cdots & x_m y_n \end{bmatrix}$$

$m \times 1$ $1 \times n$

Matrix Multiplication: Vector-Vector Products

As an example of how the outer product can be useful, let $\mathbf{1} \in \mathbb{R}^n$ denote an n -dimensional vector whose entries are all equal to 1. Furthermore, consider the matrix $A \in \mathbb{R}^{m \times n}$ whose columns are all equal to some vector $x \in \mathbb{R}^m$. Using outer products, we can represent A compactly as,

$$A = \begin{bmatrix} | & | & \cdots & | \\ x & x & \cdots & x \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} x_1 & x_1 & \cdots & x_1 \\ x_2 & x_2 & \cdots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_m & x_m & \cdots & x_m \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}}_{\mathbf{1}^T} = x \mathbf{1}^T$$

$$B = \begin{bmatrix} -y & -y & \cdots & -y \\ -y & -y & \cdots & -y \\ \vdots & \vdots & \ddots & \vdots \\ -y & -y & \cdots & -y \end{bmatrix} = \begin{bmatrix} -y \\ -y \\ \vdots \\ -y \end{bmatrix} y^T = \mathbf{1} \cdot y^T$$

Matrix Multiplication: Matrix-Vector Products

Given a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{x} \in \mathbb{R}^n$, their product is a vector $\mathbf{y} = A\mathbf{x} \in \mathbb{R}^m$.

- If we write A by rows, then we can express $A\mathbf{x}$ as,

$$\mathbf{y} = A\mathbf{x} = \begin{matrix} m \times n & n \times 1 & m \times 1 \\ \left[\begin{array}{ccc} - & \mathbf{a}_1^T & - \\ - & \mathbf{a}_2^T & - \\ & \vdots & \\ - & \mathbf{a}_m^T & - \end{array} \right] & \mathbf{x} = & \left[\begin{array}{c} \mathbf{a}_1^T \mathbf{x} \\ \mathbf{a}_2^T \mathbf{x} \\ \vdots \\ \mathbf{a}_m^T \mathbf{x} \end{array} \right] \end{matrix}.$$

In other words, the i th entry of $\mathbf{y}$ is equal to the inner product of the i th row of A and $\mathbf{x}$, $y_i = \mathbf{a}_i^T \mathbf{x}$.

Matrix Multiplication: Matrix-Vector Products

Given a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $\mathbf{x} \in \mathbb{R}^n$, their product is a vector $\mathbf{y} = A\mathbf{x} \in \mathbb{R}^m$.

- Alternatively, let's write A in column form. In this case we see that,

$$\mathbf{y} = A\mathbf{x} = \begin{bmatrix} | & | & & | \\ \mathbf{a}^1 & \mathbf{a}^2 & \cdots & \mathbf{a}^n \\ | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \overset{\text{column vector}}{=} \underbrace{[\mathbf{a}^1]} x_1 + [\mathbf{a}^2] x_2 + \cdots + [\mathbf{a}^n] x_n.$$

In other words, $\mathbf{y}$ is a linear combination of the columns of A , where the coefficients of the linear combination are given by the entries of $\mathbf{x}$.

Matrix Multiplication: Matrix-Matrix Products

Let's look at different ways of viewing the matrix-matrix multiplication
 $C = AB$.

- First, we can view matrix-matrix multiplication as a set of vector-vector products.

$$C = AB = \begin{bmatrix} - & \mathbf{a}_1^T & - \\ - & \mathbf{a}_2^T & - \\ & \vdots & \\ & \mathbf{a}_m^T & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ \mathbf{b}^1 & \mathbf{b}^2 & \dots & \mathbf{b}^p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{b}^1 & \mathbf{a}_1^T \mathbf{b}^2 & \dots & \mathbf{a}_1^T \mathbf{b}^p \\ \mathbf{a}_2^T \mathbf{b}^1 & \mathbf{a}_2^T \mathbf{b}^2 & \dots & \mathbf{a}_2^T \mathbf{b}^p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_m^T \mathbf{b}^1 & \mathbf{a}_m^T \mathbf{b}^2 & \dots & \mathbf{a}_m^T \mathbf{b}^p \end{bmatrix}$$

Matrix Multiplication: Matrix-Matrix Products

Let's look at different ways of viewing the matrix-matrix multiplication $C = AB$.

- Alternatively, we can represent A by columns, and B by rows, which leads to a much trickier interpretation of AB as a sum of outer products.

$$C = AB = \begin{bmatrix} | & | & & | \\ a^1 & a^2 & \cdots & a^n \\ | & | & & | \end{bmatrix} \begin{bmatrix} - & b_1^T & - \\ - & b_2^T & - \\ & \vdots & \\ - & b_n^T & - \end{bmatrix} = \sum_{i=1}^n a^i b_i^T$$

Matrix Multiplication: Matrix-Matrix Products

In addition to this, it is useful to know a few basic properties of matrix multiplication at a higher level:

$$\overset{m \times n}{A} \overset{n \times p}{B} \overset{p \times k}{C} \in \mathbb{R}^{m \times k}$$
$$\underline{\underline{AB}} C \in \mathbb{R}^{m \times k}$$

- Matrix multiplication is associative: $(AB)C = A(BC)$.
- Matrix multiplication is distributive: $A(B + C) = AB + AC$.
- Matrix multiplication is, in general, not commutative; that is, it can be the case that $AB \neq BA$. (For example, if $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times q}$, the matrix product BA does not even exist if m and q are not equal!)

$$A \in \mathbb{R}^{m \times n} \quad B \in \mathbb{R}^{n \times p}$$

$$C \in \mathbb{R}^{n \times n} \quad D \in \mathbb{R}^{n \times n}$$

Try to verify these properties for yourself. **Hint:** Proving matrix properties by reduction to simple scalar properties will come up often.

$$AB \in \mathbb{R}^{m \times p}$$

$$BA \in ?$$

$$CD \in \mathbb{R}^{n \times n} \stackrel{?}{=} DC \in \mathbb{R}^{n \times n}$$

Operations and Properties: Identity Matrix

The identity matrix, denoted $I \in \mathbb{R}^{n \times n}$, is a square matrix with ones on the diagonal and zeros everywhere else. That is,

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$


It has the property that for all $A \in \mathbb{R}^{m \times n}$,

$$AI = A = IA.$$

Note that in some sense, the notation for the identity matrix is ambiguous, since it does not specify the dimension of I . Generally, the dimensions of I are inferred from context so as to make matrix multiplication possible. For example, in the equation above, the I in $AI = A$ is an $n \times n$ matrix, whereas the I in $A = IA$ is an $m \times m$ matrix.

Operations and Properties: Diagonal Matrices

A diagonal matrix is a matrix where all non-diagonal elements are 0. This is typically denoted $D = \text{diag}(d_1, d_2, \dots, d_n)$, with

$$D_{ij} = \begin{cases} d_i & i = j \\ 0 & i \neq j \end{cases}$$

$$\begin{bmatrix} 1 & & \\ & 2 & \\ & & 3 \dots \end{bmatrix}$$

Clearly, $I = \text{diag}(1, 1, \dots, 1)$

I is a square 0 matrix. Diagonal. ?

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Operations and Properties: Transpose

The transpose of a matrix results from "flipping" the rows and columns. Given a matrix $A \in \mathbb{R}^{m \times n}$, its transpose, written $A^T \in \mathbb{R}^{n \times m}$, is the $n \times m$ matrix whose entries are given by

$$(A^T)_{ij} = A_{ji}.$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$$

2×3 3×2

We have in fact already been using the transpose when describing row vectors, since the transpose of a column vector is naturally a row vector.

The following properties of transposes are easily verified:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$
- $(A + B)^T = A^T + B^T$

$$A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p}$$

$$AB \in \mathbb{R}^{m \times p}$$

$$(AB)^T \in \mathbb{R}^{p \times m}$$

$$B^T \in \mathbb{R}^{p \times n}, A^T \in \mathbb{R}^{n \times m}$$

$$B^T A^T \in \mathbb{R}^{p \times m}$$

Operations and Properties: Symmetric Matrices

A square matrix $A \in \mathbb{R}^{n \times n}$ is symmetric if $A = A^T$. It is anti-symmetric if $A = -A^T$. It is easy to show that for any matrix $A \in \mathbb{R}^{n \times n}$, the matrix $A + A^T$ is symmetric and the matrix $A - A^T$ is anti-symmetric.

From this it follows that any square matrix $A \in \mathbb{R}^{n \times n}$ can be represented as a sum of a symmetric matrix and an anti-symmetric matrix, since

$$A = \frac{1}{2} (A + A^T) + \frac{1}{2} (A - A^T)$$

$$A \in \mathbb{R}^{n \times n}$$

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

$$A^T A \in \mathbb{R}^{n \times n}$$

$$A A^T \in \mathbb{R}^{n \times n}$$

$$\left[\frac{1}{2} (A + A^T) \right]^T = \frac{1}{2} (A^T + (A^T)^T) = \frac{1}{2} (A^T + A)$$

Operations and Properties: The Trace

The trace of a square matrix $A \in \mathbb{R}^{n \times n}$, denoted $\text{tr}(A)$ (or just $\text{tr} A$ if the parentheses are obviously implied), is the sum of diagonal elements in the matrix:

$$\text{tr} A = \sum_{i=1}^n A_{ii}$$

The trace has the following properties:

- For $A \in \mathbb{R}^{n \times n}$, $\text{tr}(A) = \text{tr}(A^T)$
- For $A, B \in \mathbb{R}^{n \times n}$, $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
- For $A \in \mathbb{R}^{n \times n}$, $t \in \mathbb{R}$, $\text{tr}(tA) = t \text{tr}(A)$.
- For A, B such that AB is square, $\text{tr}(AB) = \text{tr}(BA)$.
- For A, B, C such that ABC is square, $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$, and so on for the product of more matrices.

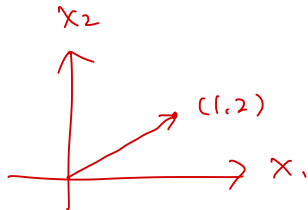
Operations and Properties: Norms

A norm of a vector $\|x\|$ is informally a measure of the "length" of the vector. For example, we have the commonly-used Euclidean or ℓ_2 norm,

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}.$$

Note that $\|x\|_2^2 = x^T x$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$



$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\|x\|_2 = \sqrt{5} = \sqrt{1^2 + 2^2}$$

$$\|x\|_2^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

Operations and Properties: Norms (Optional)

More formally, a norm is any function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies 4 properties:

- 1 For all $x \in \mathbb{R}^n$, $f(x) \geq 0$ (non-negativity).
- 2 $f(x) = 0$ if and only if $x = 0$ (definiteness).
- 3 For all $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, $f(tx) = |t|f(x)$ (homogeneity).
- 4 For all $x, y \in \mathbb{R}^n$, $f(x + y) \leq f(x) + f(y)$ (triangle inequality).

Other examples of norms are the ℓ_1 norm,

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

and the ℓ_∞ norm,

$$\|x\|_\infty = \max_i |x_i|.$$

Operations and Properties: Norms (Optional)

In fact, all three norms presented so far are examples of the family of ℓ_p norms, which are parameterized by a real number $p \geq 1$, and defined as

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

Norms can also be defined for matrices, such as the Frobenius norm,

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2} = \sqrt{\text{tr}(A^T A)}$$

Many other norms exist, but they are beyond the scope of this review.

Operations and Properties: Linear Independence

A set of vectors $\{x_1, x_2, \dots, x_n\} \subset \mathbb{R}^m$ is said to be (linearly) independent if no vector can be represented as a linear combination of the remaining vectors. Conversely, if one vector belonging to the set can be represented as a linear combination of the remaining vectors, then the vectors are said to be (linearly) dependent. That is, if

$$x_n = \sum_{i=1}^{n-1} \alpha_i x_i$$

$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$
 $n=3, m=3$

for some scalar values $\alpha_1, \dots, \alpha_{n-1} \in \mathbb{R}$, then we say that the vectors $x_1, \dots, x_n$ are linearly dependent; otherwise, the vectors are linearly independent.

$x_2 = 2x_1, \quad x_3 = 3x_1 + 0x_2$

$$\left\{ \begin{matrix} x_1 \\ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{matrix}, \begin{matrix} x_2 \\ \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \end{matrix}, \begin{matrix} x_3 \\ \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} \end{matrix} \right\}$$

Operations and Properties: Rank

The column rank of a matrix $A \in \mathbb{R}^{m \times n}$ is the size of the largest subset of columns of A that constitute a linearly independent set. In the same way, the row rank is the largest number of rows of A that constitute a linearly independent set.

For any matrix $A \in \mathbb{R}^{m \times n}$, it turns out that the column rank of A is equal to the row rank of A , and so both quantities are referred to collectively as the rank of A , denoted as $\text{rank}(A)$.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \begin{array}{l} \text{Size of set} = 3 \\ \text{Rank}(A) = 3 \end{array}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\} \quad \begin{array}{l} \text{Size of set} = 1 \\ \text{Rank}(B) = 1 \end{array}$$

Operations and Properties: Rank

The following are some basic properties of the rank:

- For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) \leq \min(m, n)$. If $\text{rank}(A) = \min(m, n)$, then A is said to be full rank.
- For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = \text{rank}(A^T)$
- (Optional) For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.
- (Optional) For $A, B \in \mathbb{R}^{m \times n}$, $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

Operations and Properties: Inverse

The inverse of a square matrix $A \in \mathbb{R}^{n \times n}$ is denoted A^{-1} , and is the unique matrix such that

$$A^{-1}A = I = AA^{-1}.$$

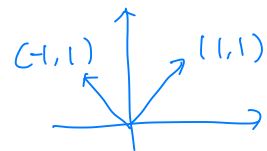
Note that not all matrices have inverses. In particular, we say that A is invertible or non singular if A^{-1} exists and non invertible or singular otherwise. In order for a square matrix A to have an inverse A^{-1} , then A must be full rank.

As an example of how the inverse is used, consider the linear system of equations, $Ax = b$ where $A \in \mathbb{R}^{n \times n}$, and $x, b \in \mathbb{R}^n$. If A is nonsingular (i.e., invertible), then $x = A^{-1}b$.

Operations and Properties: Orthogonal Matrices

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are orthogonal if $\mathbf{x}^T \mathbf{y} = 0$.
- A vector $\mathbf{x} \in \mathbb{R}^n$ is normalized if $\|\mathbf{x}\|_2 = 1$. $\sum_i x_i^2 = 1$
- A square matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal if all its columns are orthogonal to each other and are normalized (the columns are then referred to as being orthonormal).

It follows immediately from the definition of orthogonality and normality that

$$U^T U = I = U U^T.$$


$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\langle x_1, x_2 \rangle = x_1^T x_2 = [0 \ 1] \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -1 + 1 = 0$

3×3 permutation matrix

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix}$$

Operations and Properties: Orthogonal Matrices

Note that if U is not square - i.e., $U \in \mathbb{R}^{m \times n}$, $n < m$ - but its columns are still orthonormal, then $U^T U = I$, but $U U^T \neq I$.

Another nice property of orthogonal matrices is that operating on a vector with an orthogonal matrix will not change its Euclidean norm, i.e.,

$$\overset{n \times n \quad n \times 1 \Rightarrow n \times 1}{\| \underbrace{Ux}_{\text{red circle}} \|_2 = \| \underline{x} \|_2}$$

for any $x \in \mathbb{R}^n$, $U \in \mathbb{R}^{n \times n}$ orthogonal.

$$\| Ux \|_2^2 = (Ux)^T Ux = x^T \underbrace{U^T U}_I x = x^T I x = x^T x = \|x\|_2^2$$

$$z = Ux \in \mathbb{R}^n$$

$$\|z\|_2^2 = z^T z = (Ux)^T Ux = \|x\|_2^2$$

Operations and Properties: Span

The span of a set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is the set of all vectors that can be expressed as a linear combination of $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. That is,

$$\text{span}(\{\mathbf{x}_1, \dots, \mathbf{x}_n\}) = \left\{ \mathbf{v} : \mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{x}_i, \quad \alpha_i \in \mathbb{R} \right\}$$

It can be shown that if $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is a set of n linearly independent vectors, where each $\mathbf{x}_i \in \mathbb{R}^n$, then $\text{span}(\{\mathbf{x}_1, \dots, \mathbf{x}_n\}) = \mathbb{R}^n$. In other words, any vector $\mathbf{v} \in \mathbb{R}^n$ can be written as a linear combination of $\mathbf{x}_1$ through $\mathbf{x}_n$.

Operations and Properties: Range

The range (sometimes also called the columnspace) of a matrix $A \in \mathbb{R}^{m \times n}$, denoted $\mathcal{R}(A)$, is the the span of the columns of A . In other words,

$$\mathcal{R}(A) = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{v} = A\mathbf{x}, \mathbf{x} \in \mathbb{R}^n\}.$$

Operations and Properties: Nullspace

The nullspace of a matrix $A \in \mathbb{R}^{m \times n}$, denoted $\mathcal{N}(A)$ is the set of all vectors that equal 0 when multiplied by A , i.e.,

$$\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = 0\}.$$

Note that vectors in $\mathcal{R}(A)$ are of size m , while vectors in the $\mathcal{N}(A)$ are of size n , so vectors in $\mathcal{R}(A^T)$ and $\mathcal{N}(A)$ are both in $\mathbb{R}^n$.

Operations and Properties: Determinant (Reading)

Before giving the general definition for the determinant, we define, for $A \in \mathbb{R}^{n \times n}$, $A_{\setminus i, \setminus j} \in \mathbb{R}^{(n-1) \times (n-1)}$ to be the matrix that results from deleting the i th row and j th column from A . The general (recursive) formula for the determinant is

$$\begin{aligned} |A| &= \sum_{i=1}^n (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}| \quad (\text{for any } j \in 1, \dots, n) \\ &= \sum_{j=1}^n (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}| \quad (\text{for any } i \in 1, \dots, n) \end{aligned}$$

Operations and Properties: Quadratic Form

Given a square matrix $A \in \mathbb{R}^{n \times n}$ and a vector $x \in \mathbb{R}^n$, the scalar value $x^T A x$ is called a quadratic form. Written explicitly, we see that

$$x^T A x = \sum_{i=1}^n x_i (A x)_i = \sum_{i=1}^n x_i \left(\sum_{j=1}^n A_{ij} x_j \right) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

Note that,

$$= \frac{1}{2} (x^T A x + x^T A^T x)$$

$$\boxed{x^T A x} = (x^T A x)^T = \boxed{x^T A^T x} = x^T \left(\frac{1}{2} A + \frac{1}{2} A^T \right) x,$$

$$\begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & & & \vdots \\ \vdots & & & \\ A_{n1} & & & A_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} \sum_{k=1}^n A_{1k} x_k \\ \sum_{k=1}^n A_{2k} x_k \\ \vdots \\ \sum_{k=1}^n A_{nk} x_k \end{bmatrix}$$

Operations and Properties: PSD Matrices

- A symmetric matrix $A \in \mathbb{S}^n$ is positive definite (PD) if for all non-zero vectors $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^T A \mathbf{x} > 0$. This is usually denoted $A \succ 0$ (or just $A > 0$).
- A symmetric matrix $A \in \mathbb{S}^n$ is positive semidefinite (PSD) if for all non-zero vectors $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^T A \mathbf{x} \geq 0$. This is usually denoted $A \succeq 0$ (or just $A \geq 0$).

$$AA^T \quad \underline{A^T A}$$

Operations and Properties: PSD Matrices

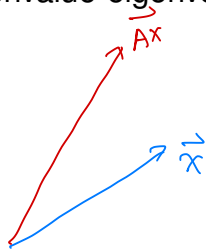
There is one type of positive definite matrix that comes up frequently, and so deserves some special mention. Given any matrix $A \in \mathbb{R}^{m \times n}$ (not necessarily symmetric or even square), the matrix $G = A^T A$ (sometimes called a Gram matrix) is always positive semidefinite.

Operations and Properties: Eigenvalues & Eigenvectors

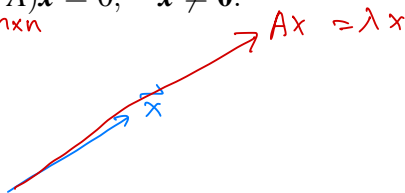
Given a square matrix $A \in \mathbb{R}^{n \times n}$, we say that $\lambda \in \mathbb{C}$ is an eigenvalue of A and $x \in \mathbb{C}^n$ is the corresponding eigenvector if

$$\begin{matrix} n \times 1 & n \times 1 \\ \underline{A}x = \lambda x, & x \neq 0. \\ n \times n & n \times 1 \end{matrix}$$

We can rewrite the equation above to state that (λ, x) is an eigenvalue-eigenvector pair of A if,



$$\begin{matrix} & n \times n \\ (\lambda I - A)x = 0, & x \neq 0. \\ n \times n & \end{matrix}$$



Operations and Properties: Eigenvalues & Eigenvectors

Remember that $(\lambda I - A)x = 0$ has a non-zero solution to x if and only if $(\lambda I - A)$ has a non-empty nullspace, which is only the case if $(\lambda I - A)$ is singular, i.e.,

$$|(\lambda I - A)| = 0.$$

We can now use the previous definition of the determinant to expand this expression $|(\lambda I - A)|$ into a (very large) polynomial in λ , where λ will have degree n . It's often called the characteristic polynomial of the matrix A .

We then find the n (possibly complex) roots of this characteristic polynomial and denote them by $\lambda_1, \dots, \lambda_n$. These are all the eigenvalues of the matrix A , but we note that they may not be distinct.

Operations and Properties: Eigenvalues & Eigenvectors

The following are properties of eigenvalues and eigenvectors (in all cases assume $A \in \mathbb{R}^{n \times n}$ has eigenvalues $\lambda_1, \dots, \lambda_n$) :

- The trace of a A is equal to the sum of its eigenvalues,

$$\text{tr } A = \sum_{i=1}^n \lambda_i$$

- The determinant of A is equal to the product of its eigenvalues,

$$|A| = \prod_{i=1}^n \lambda_i$$

- The rank of A is equal to the number of non-zero eigenvalues of A .

Operations and Properties: Eigendecomposition(1)

Assume that $A \in \mathbb{R}^{n \times n}$ is a symmetric real matrix. We have the following properties:

- All eigenvalues of A are real numbers. We denote them by $\lambda_1, \dots, \lambda_n$.
- There exists a set of eigenvectors $u_1, \dots, u_n$ such that a) for all i , u_i is an eigenvector with eigenvalue λ_i and b) $u_1, \dots, u_n$ are unit vectors and orthogonal to each other.

n eigval $\rightarrow$ n eigenvectors

Let U be the orthonormal matrix that contains u_i 's as columns: ^{*n*}

$$\underline{U} = \begin{bmatrix} | & | & \cdots & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{bmatrix}$$

$n \times n$

Operations and Properties: Eigendecomposition(2)

Let $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ be the diagonal matrix that contains $\lambda_1, \dots, \lambda_n$ as entries on the diagonal. We can verify that

$$\underline{AU = U\Lambda} \quad \checkmark \quad \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

$AUU^T = U\Lambda U^T$

Recalling that orthonormal matrix U satisfies that $UU^T = I$ and using the equation above, we have

$$\textcircled{A} = \underline{AUU^T = U\Lambda U^T}$$

Symmetric

Operations and Properties: Eigendecomposition(3)

Exercise: Assume that $A \in \mathbb{R}^{n \times n}$ is a symmetric real matrix. If the eigenvalues of A are all non negative, show that A is PSD. ($\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots \lambda_n$)

$$A = U \Lambda U^T \quad (\lambda_i \geq 0)$$

$$x \neq 0 \in \mathbb{R}^n, \quad x^T A x \geq 0$$

~~Assume~~ $x \in \mathbb{R}^n$

$$x^T A x = (x^T U) \Lambda (U^T x) = \tilde{x}^T \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \tilde{x}$$

denote $\tilde{x} = U^T x$

$$\tilde{x}^T = (U^T x)^T = x^T U$$

$$\begin{aligned} &= [\tilde{x}_1 \quad \tilde{x}_2 \quad \dots \quad \tilde{x}_n] \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_n \end{bmatrix} \\ &= [\tilde{x}_1 \quad \tilde{x}_2 \quad \dots \quad \tilde{x}_n] \begin{bmatrix} \lambda_1 \tilde{x}_1 \\ \lambda_2 \tilde{x}_2 \\ \vdots \\ \lambda_n \tilde{x}_n \end{bmatrix} = \sum_{i=1}^n \lambda_i \tilde{x}_i^2 \geq 0 \end{aligned}$$

Operations and Properties: Eigendecomposition(3)

Exercise: For a matrix $A \in \mathbb{S}^n$, consider the following maximization problem,

$$\max_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^T A \mathbf{x} \quad \text{subject to } \|\mathbf{x}\|_2^2 = 1$$

i.e., find the normalized vector which maximizes the quadratic form.

Matrix Calculus: The Gradient

Suppose that $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a function that takes as input a matrix A of size $m \times n$ and returns a real value. Then the gradient of f (with respect to $A \in \mathbb{R}^{m \times n}$) is the matrix of partial derivatives, defined as:

$$\nabla_A f(A) \in \mathbb{R}^{m \times n} = \begin{bmatrix} \frac{\partial f(A)}{\partial A_{11}} & \frac{\partial f(A)}{\partial A_{12}} & \dots & \frac{\partial f(A)}{\partial A_{1n}} \\ \frac{\partial f(A)}{\partial A_{21}} & \frac{\partial f(A)}{\partial A_{22}} & \dots & \frac{\partial f(A)}{\partial A_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(A)}{\partial A_{m1}} & \frac{\partial f(A)}{\partial A_{m2}} & \dots & \frac{\partial f(A)}{\partial A_{mn}} \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

$$\text{tr}(A) = 2 + 4 = 6$$

$$\nabla_A \text{tr}(A)$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

i.e., an $m \times n$ matrix with

$$A = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$$

$$\text{tr}(A) = x_1 + x_4$$

$$(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}$$

$$\begin{bmatrix} & \\ & \end{bmatrix}_{n \times n} \nabla_A \text{tr}(A) = I_{n \times n}$$

Matrix Calculus: The Gradient

Note that the size of $\nabla_A f(A)$ is always the same as the size of A . So if, in particular, A is just a vector $x \in \mathbb{R}^n$,

$$x \in \mathbb{R}^n$$

$$f(x) = \|x\|_2^2 = x_1^2 + x_2^2 + \dots + x_n^2 \quad \nabla_x f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}$$

$$\nabla_x f(x) = \begin{bmatrix} 2x_1 \\ 2x_2 \\ \vdots \\ 2x_n \end{bmatrix} = 2x$$

Matrix Calculus: The Hessian

Suppose that $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a function that takes a vector in $\mathbb{R}^n$ and returns a real number. Then the Hessian matrix with respect to $\mathbf{x}$, written $\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ or simply as H is the $n \times n$ matrix of partial derivatives,

Symmetric

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n} = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix}$$

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) = \begin{bmatrix} \end{bmatrix}$$

$n \times n$

$$(\nabla_{\mathbf{x}}^2 f(\mathbf{x}))_{ij} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$$

Matrix Calculus: The Hessian

In other words, $\nabla_{\mathbf{x}}^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n}$, with

$$(\nabla_{\mathbf{x}}^2 f(\mathbf{x}))_{ij} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}.$$

Note that the Hessian is always symmetric, since

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$$

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_j} \left(\frac{\partial f(\mathbf{x})}{\partial x_i} \right) = \frac{\partial}{\partial x_i} \left(\frac{\partial f(\mathbf{x})}{\partial x_j} \right),$$

Matrix Calculus: The Hessian

Note that the gradient of a vector is not defined. Therefore, it is not the case that the Hessian is the gradient of the gradient. However, this is almost true in the following sense: If we look at the i th entry of the gradient $(\nabla_{\mathbf{x}} f(\mathbf{x}))_i = \partial f(\mathbf{x}) / \partial x_i$, and take the gradient with respect to $\mathbf{x}$ we get

$$\nabla_{\mathbf{x}} \frac{\partial f(\mathbf{x})}{\partial x_i} = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_1} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_2} \\ \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_n} \end{bmatrix}$$

which is the i th column (or row) of the Hessian. Therefore,

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) = \begin{bmatrix} \nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} f(\mathbf{x}))_1 & \nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} f(\mathbf{x}))_2 & \cdots & \nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} f(\mathbf{x}))_n \end{bmatrix}.$$

Matrix Calculus: Gradient of Linear Functions

For $\mathbf{x} \in \mathbb{R}^n$, let $f(\mathbf{x}) = \mathbf{b}^T \mathbf{x}$ for some known vector $\mathbf{b} \in \mathbb{R}^n$. Then

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_i} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \mathbf{b}$$

$f(\mathbf{x}) = \sum_{i=1}^n b_i x_i$

$f(\mathbf{x}) = \mathbf{b}^T \mathbf{x} = [b_1 \ b_2 \ \dots \ b_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$= \sum_{i=1}^n b_i x_i$

Matrix Calculus: Gradient of Quadratic Functions

The quadratic function $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ for $A \in \mathbb{S}^n$:

$$f(\mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j \quad \text{CS 229}$$

Matrix Calculus: Hessian of Linear Functions

For $\mathbf{x} \in \mathbb{R}^n$, let $f(\mathbf{x}) = \mathbf{b}^T \mathbf{x}$ for some known vector $\mathbf{b} \in \mathbb{R}^n$. Then

$$\mathbb{R}^n \rightarrow \mathbb{R}$$

$$f(\mathbf{x}) = \sum_{i=1}^n b_i x_i \quad \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_i} \right) = 0$$

$\underline{\underline{b_i}}$

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) = H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & & \\ & \text{[circle]} & \\ & & \end{bmatrix} \quad \begin{matrix} n \\ n \end{matrix}$$

$$\frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = 0$$

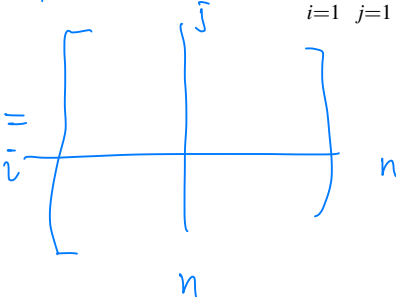
$\underline{\underline{b_j}}$

Matrix Calculus: Hessian of Quadratic Functions

The quadratic function $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ for $\mathbf{A} \in \mathbb{S}^n$:

$\mathbf{x} \in \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(\mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) =$ 

$\frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right)$

Matrix Calculus: Least Squares

To recap,

- $\nabla_{\mathbf{x}} \mathbf{b}^T \mathbf{x} = \mathbf{b}$
- $\nabla_{\mathbf{x}} \mathbf{x}^T \mathbf{A} \mathbf{x} = 2\mathbf{A} \mathbf{x}$ (if $\mathbf{A}$ symmetric)
- $\nabla_{\mathbf{x}}^2 \mathbf{x}^T \mathbf{A} \mathbf{x} = 2\mathbf{A}$ (if $\mathbf{A}$ symmetric)

Matrix Calculus: Least Squares

Suppose we are given matrices $A \in \mathbb{R}^{m \times n}$ (for simplicity we assume A is full rank) and a vector $\mathbf{b} \in \mathbb{R}^m$ such that $\mathbf{b} \notin \mathcal{R}(A)$. In this situation we will not be able to find a vector $\mathbf{x} \in \mathbb{R}^n$, such that $A\mathbf{x} = \mathbf{b}$, so instead we want to find a vector $\mathbf{x}$ such that $A\mathbf{x}$ is as close as possible to $\mathbf{b}$, as measured by the square of the Euclidean norm $\|A\mathbf{x} - \mathbf{b}\|_2^2$.

Matrix Calculus: Least Squares

Using the fact that $\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$, we have

$$\begin{aligned}\|A\mathbf{x} - \mathbf{b}\|_2^2 &= (A\mathbf{x} - \mathbf{b})^T (A\mathbf{x} - \mathbf{b}) \\ &= \mathbf{x}^T A^T A \mathbf{x} - 2\mathbf{b}^T A \mathbf{x} + \mathbf{b}^T \mathbf{b}\end{aligned}$$

Questions

Questions